

# Learning Report

## 1. Why ViT?

- **Global Context via Self-Attention:** Unlike CNNs that process locally, ViTs use self-attention to relate every image patch simultaneously from the first layer. This captures long range dependencies necessary for analyzing complex structures like galaxy morphologies.
- **Reduced Inductive Bias:** CNNs enforce strict biases like translation invariance and locality. ViTs start with minimal assumptions, relying on data to learn relationships and uncover deeper, non-local patterns that rigid convolutions might miss.
- **Exceptional Scalability:** ViTs scale effectively with larger model sizes and massive datasets without plateauing as early as CNNs, making them the standard for powerful foundation models like DINOv2.
- **Token-Based Representation:** Processing image patches like text sequences yields high quality semantic embeddings. This allows straightforward extraction of patch tokens for local features and the CLS token for global representations, which is essential for DESI dataset inference.

## 2. How Image Patchification Works?

Image patchification transforms a 2D image into a sequential 1D token list, enabling Transformers to process it like text.

Here is the technical sequence of how it works:

- **Initial State:** The image loads as an  $H \times W \times C$  (Height, Width, Channels) 3D tensor. For the DESI dataset, this is  $160 \times 160 \times 4$  for the g, r, i, and z astronomical bands.
- **Grid Slicing:** The tensor is divided into non-overlapping  $P \times P$  patches (typically  $16 \times 16$ ), yielding  $(H \times W) / (P \times P)$  total blocks.
- **Flattening:** Each  $P \times P \times C$  patch is unrolled into a flat 1D vector.
- **Linear Projection:** A trainable linear layer projects each flattened vector into a uniform embedding dimension (D) for computational efficiency.
- **Final Output:** The original 3D tensor becomes an  $N \times D$  (Number of Patches  $\times$  Embedding Dimension) 2D matrix. This sequence serves as the patch embedding input for the Transformer.

## 3. Patch Embeddings

- The continuous vector representations generated by flattening image patches and passing them through a linear projection layer. They convert raw pixel chunks into the tokens that the Transformer architecture can process.

## 4. Positional Embeddings

- Learnable vectors added directly to the patch embeddings to retain spatial information. Because Transformers process all tokens simultaneously and have no inherent sense of order, these embeddings act as coordinates, telling the model the original 2D location of each patch within the image.

## 5.CLS token

- A special, learnable vector added to the very beginning of the patch embedding sequence. As it passes through the Transformer layers, the self attention mechanism allows it to aggregate information from every other patch. By the final layer, the CLS token outputs a compressed, global summary representation of the entire image, making it ideal for classification or whole image analysis.

## 6.ViT Transformer Architecture

**Input Sequence:** Combines the flattened patch embeddings, positional embeddings, and the CLS token into a single 1D sequence.

**Transformer Encoder Blocks:** The core processing engine. The sequence passes through multiple identical layers, each containing:

- **Multi-Head Self-Attention (MHSA):** The mechanism that allows every patch token to dynamically share and gather information from all other patches globally.
- **Multi-Layer Perceptron (MLP):** A feed-forward network that processes each token individually to extract deeper patterns.
- **LayerNorm & Skip Connections:** Normalization and residual connections are applied around the MHSA and MLP to stabilize training and ensure smooth information flow across deep layers.

**Output:** The network outputs fully updated tokens. The final CLS token is extracted as a global summary of the image, while the final patch tokens are extracted for localized, spatial features.

## 7.CLS vs patch tokens

- **CLS Token:** A single, global summary token that gathers context from the entire image. It is used to understand the image as a whole, such as determining its overall classification.
- **Patch Tokens:** Multiple, localized tokens that each represent one specific physical square of the original image. They are used to analyze specific spatial details, structures, and locations within the image.

## 8.DINOv2

**DINOv2** is a powerful, self-supervised Vision Transformer (ViT) foundation model developed by Meta. Here is what makes it unique:

- **No Labels Needed:** Unlike traditional models, it trains on massive datasets of raw images without needing any human-provided tags or categories (like "cat" or "galaxy").
- **Self-Taught:** It teaches itself to understand visual structures by comparing different cropped views of the same image, learning deep, universal patterns on its own.
- **Universal Features:** Because it isn't restricted to specific labels during training, it produces incredibly rich, ready-to-use embeddings. Its output tokens (both global CLS

and local patch tokens) can be plugged directly into almost any vision task with high performance right out of the box.

## 9.DINOv2 Outputs

When an image is passed through DINOv2, it outputs a complex tensor that contains two primary representations:

- **One Global CLS Token:** A single, high dimensional vector that serves as a comprehensive, semantic summary of the entire image. This is used for whole image tasks like classification or similarity matching.
- **A Sequence of Local Patch Tokens:** A collection of vectors where each vector corresponds to one specific physical square (usually 14x14 or 16x16 pixels) of the original image. Because these retain their spatial locations, they are used for dense, detail oriented tasks like segmentation or identifying specific structures within a galaxy.

## 9.How DINOv2 can be used in this project

- **Instant Feature Extraction:** Instead of training a model from scratch, DINOv2 quickly processes the 160×160 g/r/i/z DESI images into ready-to-use mathematical representations.
- **Global Summary (CLS Token):** This token provides a compressed snapshot of the entire galaxy, which you will use to understand overall classifications like the Galaxy Zoo morphology labels.
- **Local Details (Patch Tokens):** These tokens map to specific grid locations, allowing you to inspect the spatial dimensions and underlying structures of the galaxies.